

Chapter 150

Using LUTs

Lookup Tables, also known as LUTs, are one of the most ubiquitous means of creating, exchanging, and applying image processing operations there is for purposes of color management, display calibration, look management, and general-purpose processing of image color and contrast. DaVinci Resolve has robust support for LUTs throughout its image processing pipeline.

Contents

What is a LUT?	3406
Supported LUT Formats	3407
LUTs and ACES	3408
Adding Lookup Tables of Your Own	3408
Custom LUT Paths	3408
LUT Paths for the macOS App Store Version of DaVinci Resolve	3409
LUT Controls in the Project Settings	3409
Applying LUTs to Source Clips	3409
Using the Color Page LUT Browser	3410
Applying a LUT Within a Node	3412
LUTs Are the Last Operation Within a Node	3412
Favorite LUTs Submenu in Node Editor	3413
Missing LUTs	3413
Exporting LUTs	3413

What is a LUT?

LUTs are simply files, similar to plugins but far more focused and with no user interface, that specify image processing operations. These operations are accomplished in a variety of ways. The traditional approach is to use a 1D table or 3D “cube” of pre-calculated values to perform an image color transform. However, newer LUT formats including CLF and DCTL let you use mathematical scripts to process an image.

Whatever type of LUT you use, these files can be loaded into DaVinci Resolve and applied at different points of the image processing pipeline to apply image processing operations for different purposes. There are several well-known uses of LUTs, but the important thing you should take away is that LUTs are simply color transform operations that can be used for many things, and there’s no single use of a LUT that’s more or less important than any other.

Here are some frequent uses for LUTs:

- While optionally superseded by Resolve Color Management (RCM), lookup tables (LUTs) have been frequently used to create a starting point adjustment for media acquired with some logarithmic encoding. DPX log film scans, digital media using the ARRI ALEXA’s Log-C encoding, Sony’s S-Log exposure setting, or RED R3D media that is debayered using the REDFilmLog setting are all examples of media using a logarithmic exposure curve, designed to protect as much detail in the highlights and shadows of a digitally encoded image as possible. While log-encoded media retains a lot of image data, the picture is initially flat and unsuitable for use without grading. The exposure and color must be adjusted to “normalize” the media, making it look closer to the way it’s supposed to, in order to start grading. While you can do this manually, it’s usually faster to use a LUT that’s tailored to your type of media and the exposures you’re using. Alternately, you can also use Resolve Color Management to accomplish this.
- LUTs are commonly used in onset workflows where dailies for different scenes are managed with corresponding LUTs. These LUTs were used to monitor the media as it was being recorded to define a baseline reference for how each scene is meant to look, at least so far as field monitoring is concerned. In more advanced workflows, LUTs are used as a baseline look, defined prior to the shoot and used during the shoot, that then defines the creative starting point for different scenes once grading begins after the shoot.
- LUTs are frequently used as a stylistic component of a grade, or “look” that gives users a quick start when desiring some manner of creative adjustment. Over the years, companies and individuals have created an ecosystem of such looks that are disseminated and sold in various LUT formats supported by DaVinci Resolve.

In all of these instances, LUTs are simply image processing adjustments that are applied to affect the color and contrast of a clip, in much the same way as you’d make adjustments using any of the contrast or color controls in the Color page.

Supported LUT Formats

DaVinci Resolve uses both 1D and 3D LUTs, and supports LUTs in a variety of formats.

- **.cube:** DaVinci Resolve uses both 1D and 3D LUTs in the .cube format. 3D LUTs can be exported as 17x17x17, 33x33x33, or 65x65x65 cubes with 32-bit floating point processing. DaVinci Resolve can also read and use Shaper LUTs in the .cube format, but these must also be created outside of DaVinci Resolve. You should note that while 17-point LUTs are not recommended to use for grading, they are useful when exporting LUTs for on-set monitoring, to accommodate different display, calibration, and signal conversion devices.
- **Panasonic VLUT format:** DaVinci Resolve can both read and generate this LUT format, designed for use in the Panasonic VariCam camera ecosystem.
- **Video Range LUTs:** Beginning with DaVinci Resolve 17, support was added for importing LUTs that include additional metadata specifying them as processing image data in Video Range, rather than Full Range. More information about how to format LUTs as Video Range is included in the Developer Documentation available from the Help menu. Being able to specify whether a LUT is meant to be Video or Full Range allows LUT processing to automatically accommodate DaVinci Resolve's data range setting pipeline for clip attributes, project settings, and output settings.
- **CLF (common LUT format):** CLFs use an XML format that is capable of encompassing a limited number of mathematical transforms in addition to traditional lookup-tables to do image processing. Promoted by the academy as the ideal LUT format for use with ACES, LMTs for ACES are recommended to be in the CLF format due to its increased precision and flexibility.
- **DCTL:** DCTL files are actually color transformation scripts that DaVinci Resolve sees and applies just like any other LUT. Unlike other LUTs, which are 1D or 3D lookup tables of values that approximate image transformations using interpolation, DCTL files are actually comprised of computer code that directly transforms images using combinations of math functions that you devise. Additionally, DCTL files run natively on the GPU of your workstation, so they can be fast.

For more information on DCTL, see *Chapter 200, "Creating DCTL LUTs."*

What's the Difference Between a LUT and a Shaper LUT?

DaVinci Resolve is capable of importing and using LUTs within its 32-bit floating point image processing pipeline. The .cube format can be used as either a simple 33x33x33 3D LUT, or as a shaper LUT, which is actually a method of using two LUTs, a 1D LUT and a 3D LUT together, that addresses signal processing issues that 3D LUTs alone can't handle.

For processor efficiency, 3D LUTs are designed with reasonable lower and upper limits for the data they will handle. It's well known that when a 3D LUT is fed values that are outside of the range that LUT is designed to handle, the out-of-range data will be clipped. Since many LUTs are designed with digital cinema workflows in mind, the practical result is that feeding a video signal with super-white in it to a 3D LUT that's designed for full-range data (0–1) will clip the super-white part of the signal.

Shaper LUTs handle this issue by first using a 1D LUT to process video signals with out-of-range data, fitting the signal into a range that the 3D LUT won't clamp. The output of the 3D LUT includes the reverse transformation, to effectively zero out the 1D LUTs transform, while retaining whatever processing the 3D LUT was meant to apply.

Shaper LUTs are also useful for dealing with extremely large data sets, such as OpenEXR files that can theoretically handle an image data range of $-\infty$ to $+\infty$. Using a Shaper LUT, you can remap the incoming data to fit more precision in the 0–1 range, leaving less important data outside the range.

LUTs and ACES

The academy that promotes the correct use of ACES strongly recommends that LUTs be processed in the ACES color space. For this reason, two project settings let you choose how this will be done:

- **ACEScc AP1 Timeline Space:** This setting works for either the ACEScc or ACEScct Color Science settings depending on what you've selected in the Color Science pop-up at the top of the Color Management panel in the Project Settings. This setting lets you use LUTs that were created for ACES workflows that are similar to (but not the same as) LUTs you would create and use in a traditional log-encoded workflow. LUTs designed for working with this setting should have a range of -0.358 to 1.468 so that grading operations that clip an image from 0 to 1 won't destroy the look being applied.

The “ACEScc AP1 Timeline Space” setting is also good for workflows where you want to use conventional LUTs that were designed for Rec. 709 workflows using the ResolveFX “ACES Transform” plugin, which over a series of three nodes lets you transform the image from ACES to 709, apply a Rec. 709-designed LUT, and then convert the image from 709 back to ACES.

- **ACES APO Linear:** This setting requires you to apply an LMT LUT that has been specifically created for ACES image data. Only use this setting if you're using CLFs that are designed for ACES using a range of -65504 to 65504 , as specified by SMPTE 2065.

Adding Lookup Tables of Your Own

The menus in the Color Management panel of the Project settings include a series of factory preset LUTs that were installed with DaVinci Resolve, along with any LUTs that have been generated by DaVinci Resolve, or that you've imported into the proper directory for your operating system for your own use.

By default, LUTs are saved to the following locations:

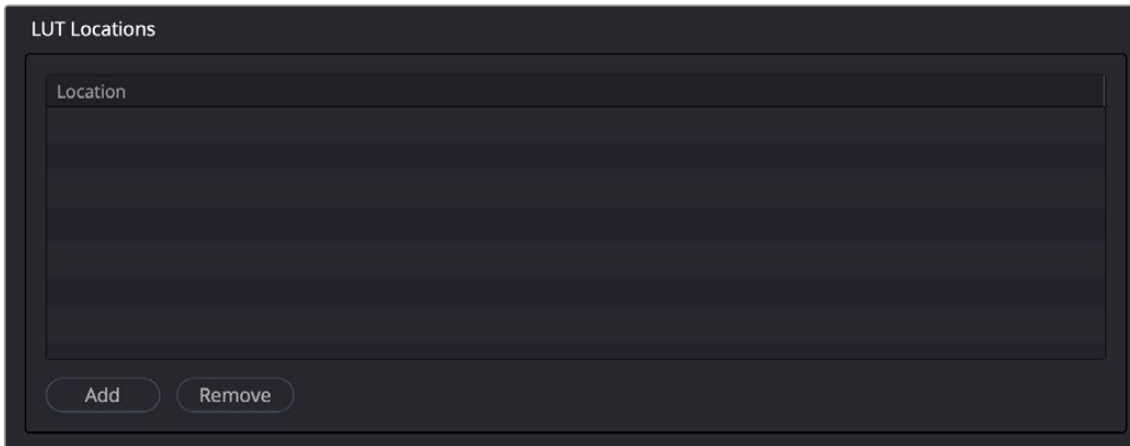
On OS X: Library/Application Support/Blackmagic Design/DaVinci Resolve/LUT/

On Windows: C:\ProgramData\Blackmagic Design\DaVinci Resolve\Support\LUT

On Linux: /home/resolve/LUT

Custom LUT Paths

A list in the General panel of the DaVinci Resolve System Preferences lets you add multiple file paths for loading LUTs that you want to use in DaVinci Resolve. This works for network volumes for facilities where multiple workstations are accessing a central collection of LUTs to be shared among multiple artists. Clicking the Add button lets you add a file path to this table from a dialog. Selecting a location in this list and clicking Remove removes that location.



A list in the General panel of the System Preferences lets you add multiple locations where LUTs you want to use from within DaVinci Resolve are located.

LUT Paths for the macOS App Store Version of DaVinci Resolve

If you downloaded the non-studio version of DaVinci Resolve from the Apple App Store, LUTs are saved in a different location in order for DaVinci Resolve to remain totally self-contained. In this case, you can click the Open LUT Folder button in the Lookup Tables panel of the Project Settings, to open up a Finder window at the location these LUTs are stored. You can use this window to copy LUTs that you want DaVinci Resolve to have access to, or delete LUTs that you no longer need.

If you add a LUT to one of these directories after DaVinci Resolve has been opened, you can click the Update Lists button to refresh the contents of the pop-up menus.

LUT Controls in the Project Settings

While there's an entire group of LUT controls in the Color Management panel of the Project Settings, those are designed to apply LUTs, at different parts of the image processing pipeline, to the entire Timeline. This is useful when you want to apply a single color and contrast transformation to the entire program at once, but less so if you want to apply different LUTs on a per clip basis.

For more information on using the Lookup Table settings, see *Chapter 4, "System and User Preferences."*

Applying LUTs to Source Clips

Another way of applying a LUT to a clip is to apply it directly to the source clip, which you can do to any clip in the Media Pool, or in the Thumbnail Timeline of the Color page. This can be convenient, but keep in mind that source clip LUTs cannot be copied from one timeline to another using ColorTrace, so using source clip LUTs limits your potential workflows. For most workflows, it's better to apply LUTs directly in the Node Editor so they live in each clip's grade.

To apply a LUT to one or more selected clips in the Media Pool:

- Right-click one of the selected clips, and choose a LUT from the 1D LUT or 3D LUT submenus.

To apply a LUT to one or more selected clips in the Thumbnail Timeline of the Color page:

- Right-click one of the selected thumbnails, and choose a LUT from the 1D LUT or 3D LUT submenus.

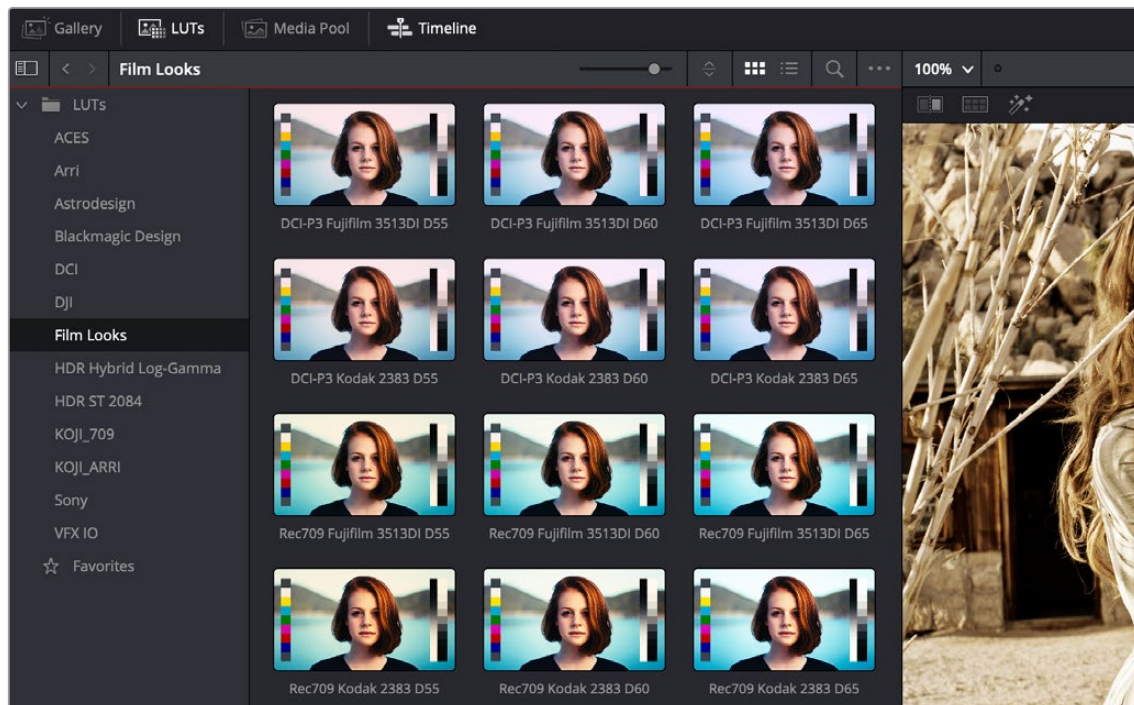
TIP: If you're wanting to apply an image transformation to normalize log-encoded source clips, you may also consider using Resolve Color Management (RCM). Depending on your source media, it may be an easier and more automated process.

For more information, see *Chapter 9, "Data Levels, Color Management, and ACES."*

Using the Color Page LUT Browser

The LUT Browser on the Color page provides a centralized area for browsing and previewing all of the LUTs installed on your workstation. All LUTs appear in the sidebar, by category.

By default, all LUTs appear with a test thumbnail that give a preview of that LUT's effect, but you can also get a live preview of how the current clip looks with that LUT by hover scrubbing with the pointer over a particular LUT's thumbnail (this is described in more detail below).



The LUT Browser

To open the LUT Browser:

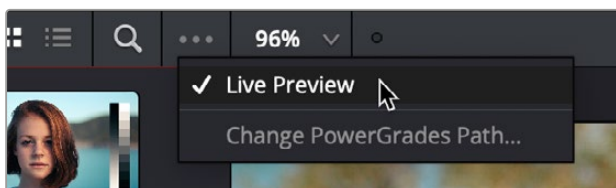
- Click the LUT Browser button in the UI Toolbar at the top of the Color page.

Methods of working with the LUT Browser:

- **To see the LUTs in any category:** Click on a LUT category to select it in the sidebar, and its LUTs will appear in the browser area.
- **To make a LUT a favorite:** Hover the mouse over a LUT and click the star badge that appears at the upper right-hand corner, or right-click any LUT and choose Add to Favorites. That LUT will then appear when you select the Favorites category in the Node Editor contextual menu for nodes.
- **To search or filter for specific LUTs:** Open a bin that has the LUT you're looking for, then click the magnifying glass icon to open the search field, and type text that will identify the LUTs you're looking for.
- **To see LUTs in Column or Thumbnail view:** Click the Column or Thumbnail buttons at the top right of the LUT Browser to choose how to view LUTs in the browser area.
- **To sort LUTs in Thumbnail view:** Click the Thumbnail Sort pop-up menu and choose which criteria you want to sort LUTs by. The options are filename, type, relative path, file path, usage, date modified. There are also options for ascending and descending sort modes.
- **To sort LUTs in Column view:** Click the column header to sort by that column. Click a header repeatedly to toggle between ascending and descending modes.
- **To update the thumbnail of a LUT with an image from a clip:** Choose a clip and frame that you want to use as the new thumbnail for a particular LUT, then right-click that LUT and choose Update Thumbnail With Timeline Frame.
- **To reset the thumbnail of a LUT to use the standard thumbnail:** Right-click a LUT and choose Reset Thumbnail to go back to using the standard test image.
- **To refresh a LUT category with new LUTs that may have been installed:** Select a LUT category, then right-click anywhere within the browser area and choose Refresh to refresh the contents of that category from disk.

Enabling and Disabling the LUT Viewer live preview:

- 1 Open the LUT Viewer's option menu and choose Live Preview.



The Live Preview option for the LUT browser lets you hover over a LUT to preview it on the current clip in the Viewer

- 2 Click a node in the Node Editor you want to preview applying a LUT to. The live preview will display how the current clip will appear with the LUT you select applied to the currently selected node of the current grade, which will affect the result.
- 3 Move the pointer over the LUT you want to preview.

The Viewer image updates to show how that clip would look with that LUT applied to the currently selected node.

To apply a LUT from the LUT Browser to a specific node, do one of the following:

- Right-click a LUT and choose Apply LUT to current node.
- Drag a LUT from the LUT Browser and drop it onto the node you want to apply a LUT to. If you drag a LUT onto a node that already has a LUT, the previous LUT will be overwritten by the new one.

Applying a LUT Within a Node

DaVinci Resolve lets you apply LUTs within a grade by connecting a LUT to a particular node in the Node Editor. This gives you the greatest amount of control over where the LUT is applied in your image processing pipeline, and it also gives you the opportunity to apply image adjustments prior to the LUT, and after the LUT, as you require.

To apply a LUT within a node, do one of the following:

- Right-click any node, and choose a LUT from the 1D Input LUT, 1D Output LUT, 3D LUT, DCTL, or CLF (common LUT format) submenus of the LUT submenu. The LUT submenus list whichever LUTs have been installed on your workstation.

For more information on installing LUTs, see *Chapter 4, "System and User Preferences."*

- Right-click any node, and choose a LUT from the LUT > Favorites submenu.
- Use the LUT Browser to find a LUT you want to use, and then drag and drop that LUT onto the node you want to apply it to

TIP: If you hold down the Option key while scrolling through the submenu of LUTs in a Corrector node's contextual menu, you'll get a live update in the Viewer of how each LUT affects the image.

To reveal a selected node's LUT:

For any node in the Node Editor with a LUT applied to it, you can right-click that node and choose Reveal Selected LUT to automatically open the LUT viewer and select that LUT.

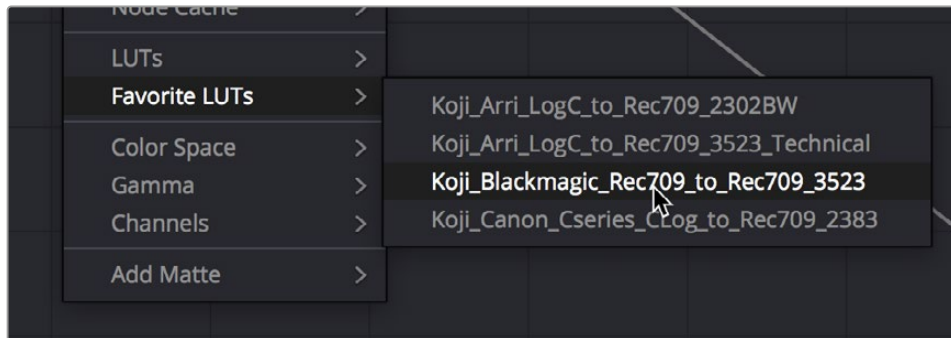
LUTs Are the Last Operation Within a Node

Each node in the Node Editor is capable of performing multiple operations, and these operations occur in a specific order. LUTs that you add to a node impose their transformation as the last operation within that node, after all other Color page adjustments applied by that node.

Practically, this means that you can use a node's Color and Contrast controls to trim the image data that will be fed into a LUT applied to that same node. For example, if a LUT's contrast adjustment clips the highlights of the image too much, you can use that node's Contrast controls to lower the highlights of the image prior to the LUT, restoring detail to the image.

Favorite LUTs Submenu in Node Editor

When you “star” a LUT as a favorite in the LUT Browser, those favorite LUTs appear in a submenu of the contextual menu that appears when you right-click on a node in the Node Editor. This makes it easy to create a short list of your go-to LUTs for various situations, for rapid application right in the Node Editor.



A Favorite LUTs submenu in the Node Editor contextual menu gives you a short list

Missing LUTs

Clips with missing LUTs show an overlay at the bottom right of the screen indicating the name of the LUT if a single LUT is missing, or an indicator that multiple LUTs are missing. This allows for quick previews of missing LUTs without interruption. Multiple missing LUTs can be seen and managed from the Missing LUTs tab in the LUT Browser, which only appears if you have one or more LUTs missing.

Exporting LUTs

If you find it necessary to exchange image adjustments with other grading applications, compositing applications, or NLEs, often the easiest inter-application solution is to export a LUT. This can be done whether your grade consists of one node or several nodes, so long as they contain only Primaries palette adjustments, Custom Curves palette adjustments, and compatible ResolveFX plugins that include Color Space Transform, ACES Transform, and Gamut Mapping. All nodes with compatible functions will be mathematically combined and translated into a LUT.

Keep in mind that any nodes that use Qualifiers, Windows, incompatible filtering operations (such as sharpening or blurring), or incompatible Resolve FX or Open FX will be completely ignored, as will all other correction operations made within these nodes.

To export a LUT:

- 1 Right-click a clip thumbnail in the Timeline of the Color page, and choose an option from the Generate LUT submenu:
 - **Generate 3D LUT (17 Point Cube):** A DaVinci-developed LUT format
 - **Generate 3D LUT (33 Point Cube):** A DaVinci-developed LUT format
 - **Generate 3D LUT (65 Point Cube):** A DaVinci-developed LUT format
 - **Generate 3D LUT (Panasonic VLUT):** A LUT format associated with Panasonic VariCam cameras

- 2 Choose a location for the resulting LUT file in the file dialog. The default file path depends on your operating system; saving it here will make it available to DaVinci Resolve for future use.

On OS X: Library/Application Support/Blackmagic Design/DaVinci Resolve/LUT/

On Windows: C:\ProgramData\Blackmagic Design\DaVinci Resolve\Support\LUT

On Linux: /home/resolve/LUT

If you like, you can create a new folder in which to save your custom LUTs.

- 3 Enter a name into the Save As field, and click Save. A LUT file is saved.

Once created, you can use an exported LUT from within DaVinci Resolve, applying it to a clip or node directly, or applying it to the entire project using the settings found in the Color Management panel of the Project Settings. You can also copy the LUT to a memory stick to give to someone for monitoring or previewing in an onset workflow, or to someone using another grading application that can read this LUT format.



Color Page Effects

CONTENTS

151	Using Open FX and Resolve FX.....	3416
152	Sizing and Image Stabilization	3429
153	The Motion Effects and Blur Palettes.....	3445